Multivariate Data Analysis coursework

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This report explores the properties of simple kernal regression models in Python. We illustrate optimal prediction, confidence bands, and conditional realisations for one-dimensional data using the Gaussian kernel and three other kernels. Maximum-likelihood estimation of the kernel parameter ρ is performed, and its sampling distribution is examined. We then apply this method to our 50-point `DataVolcano50` dataset, producing a predicted surface and confidence intervals. Finally, we extend the model to include Gaussian observation noise and finite-dimensional polynomial kernels.

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1 Introduction

Kernel regression provides us with a flexible non-parametric way to predict a function $z : X \to \mathbb{R}$ from noisy or noise-free observations. Given a kernel function $K(x, x')$ and a set of sample locations $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathcal{X}$,which are used to make the symmetric and positive-semidefinite kernel matrix K with corresponding responses $\mathbf{z} \in \mathbb{R}^n$, the optimal predictor at a new point x is

$$\hat{z}(x) = \mathbf{k}(x)^\top \mathbf{K}^{-1}\mathbf{z},$$

where $\mathbf{k}(x) = (K(x, \mathbf{x}_1), \ldots, K(x, \mathbf{x}_n))^\top$ and $\mathbf{K}$ is the kernel matrix evaluated at the sample locations. The conditional-covariance is

$$c - cov(Z_x, Z_{x'}|z) = K(x, x') - \mathbf{k}(x)^\top \mathbf{K}^{-1}\mathbf{k}(x').$$

This report reproduces and extends the experiments contained in the supplied Jupyter notebook. We first study the one-dimensional case, compare 3 different kernels, estimate the ρ parameter via maximum likelihood estimation, and then apply the method to the three-dimensional volcano dataset using a two dimensional kernel. In addition we investigate other ways of estimating ρ, models with observation noise and kernel regression with finite dimensional kernels. In this report we were able to see the effect that using different kernel functions and different values of ρ has on the result of the kernel regression.

2 One-dimensional Kernel Regression

2.1 Gaussian kernel and basic predictions

We discretised the interval $[0, 10]$ with 501 points and simulated a centred Gaussian process with Gaussian (squared-exponential) kernel

$$K(x, x') = \exp(-\rho\|x - x'\|^2), \qquad \rho = 1.$$

Seven observation locations were selected and the corresponding response vector was extracted from one prior realisation (seed 33). The optimal predictor, 95% confidence band .

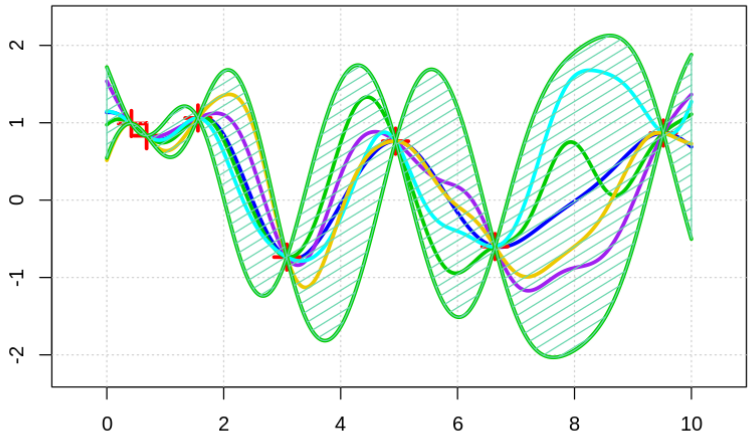
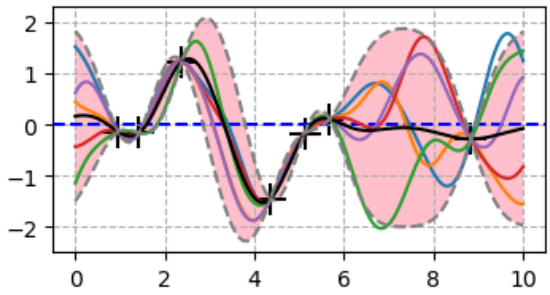


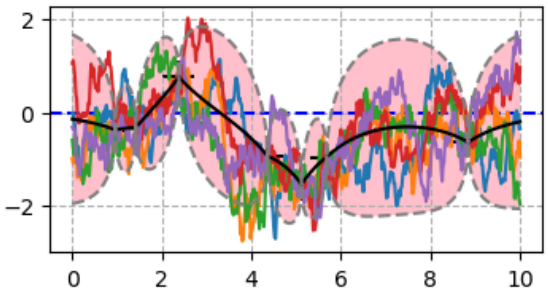
Figure 1: The optimal prediction(using a Gaussian kernel with $\rho = 1$) is the light blue line together with 4 conditional realisations and the green region is the 95% confidence band.

2.2 Effect of kernel choice

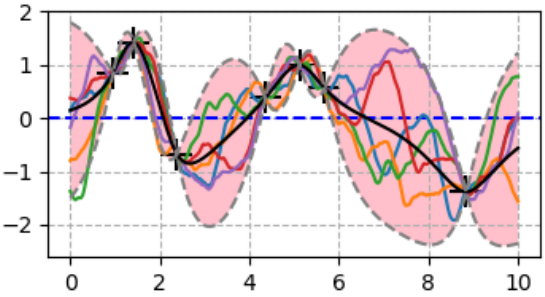
We then repeated the analysis with three additional kernels Laplace and Matérn-3/2 using the same parameter $\rho = 1$. Figure 2 shows that the Gaussian and Matérn-3/2 kernels produce visibly smoother predictions to that of the Laplace kernel, which exhibits sharper changes near observations. This confirms the well-known smoothness properties encoded by the kernel: the Gaussian kernel yields infinitely differentiable sample paths, whereas the Laplace kernel yields non-differentiable paths.



(a) Gaussian



(b) Laplace



(c) Matérn-3/2

Figure 2: The black lines are the optimal predictions, black marks represent the 7 observations the grey dashed lines are the upper and lower 95% confidence intervals, with the pink area being the confidence region and the multi coloured lines are the conditional realisations.

In figure 2 we have the comparison of optimal predictions obtained with three different kernels ($\rho = 1$) on the same one-dimensional dataset. The Gaussian kernel is the smoothest; the Laplace kernel is the roughest.

2.3 Maximum-likelihood estimation of ρ

Treating ρ as unknown, we maximized the Likelihood of the seven observations under the Gaussian kernel using the likelihood function $L(\rho|\mathbf{z} = z) = \dfrac{1}{Det(2\pi K_p)} e^{-\frac{1}{2} z^T K_P^{-1} z}$. The Likelihood (Figure 3) is unimodal and attains its maximum at $\rho_{\text{ML}} \approx 1.1384$ (exact value obtained from using the match function).

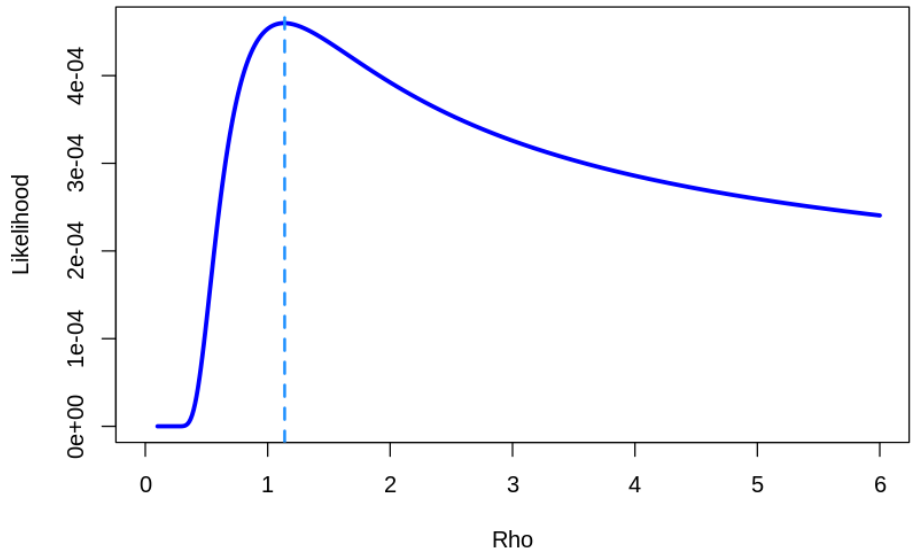
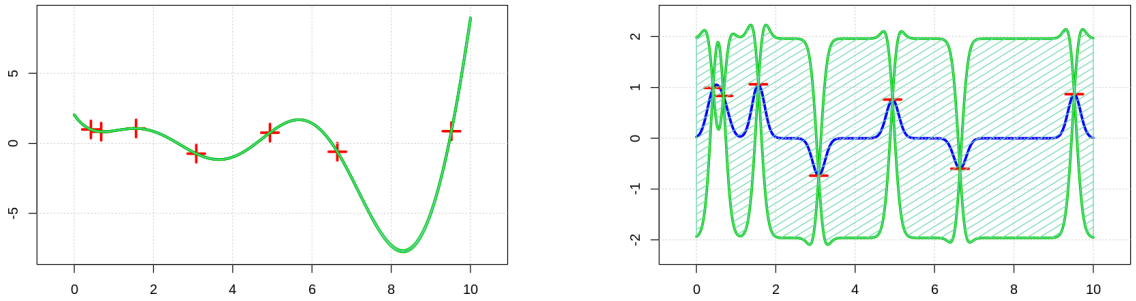


Figure 3: Likelihood as a function of ρ (Gaussian kernel). The vertical line marks the maximum-likelihood estimate ρ_{ML}.

We also illustrated under- and over-fitting by using $\rho = 20$ and $\rho = 0.05$ alongside the ML value .



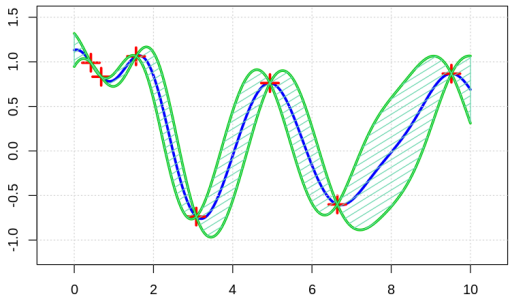
(a) Prediction made using a gaussian kernel with $\rho = 0.05$

(b) Prediction made using a gaussian kernel with $\rho = 20$

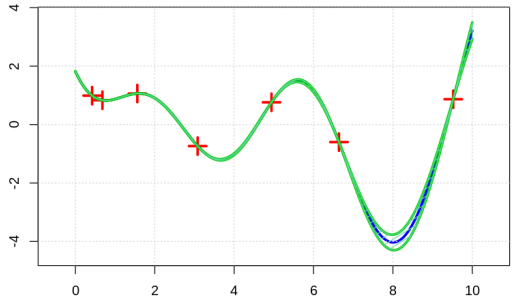
Figure 4

2.4 Adding different parameters to the gaussian kernel

You can also add different parameters to the kernel. For example, we tried adding a scalar to the kernel so that the new kernel is defined by $K(x, x') = \alpha \exp(-\rho \|x - x'\|^2)$. Changing this parameter at first sight seemed to have a similar, albeit much less powerful, effect to changing the ρ parameter. This is showcased in the figures below.



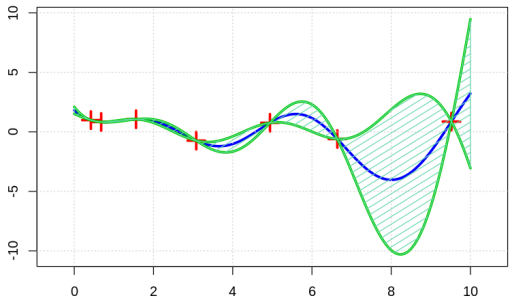
(a) Optimal prediction using gaussian kernel with $\alpha = 0.1$(true value = 1)



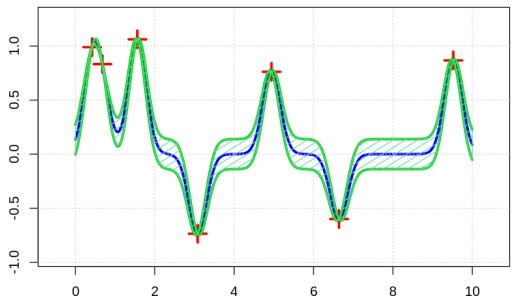
(b) Optimal prediction using gaussian kernel with $\rho = 0.1$(true value = 1)

Figure 5: Optimal predictions with varying α and ρ parameters.

Seeing this we experimented with trying to correct a large ρ parameter by using a small α. However after the experiments we found that α and ρ didn't have the same effect on the prediction.



(a) Optimal prediction using gaussian kernel with $\rho = 0.1$(true value = 1) and $\alpha = 5000$(true value = 1)



(b) Optimal prediction using gaussian kernel with $\rho = 10$(true value = 1) and $\alpha = 0.005$(true value = 1)

Figure 6: Optimal predictions with varying α and ρ parameters.

This has led us to believe that α primarily has an effect on the confidence interval as opposed to the prediction as when you increase α the CI widens and when you decrease α the CI gets thinner.

To further investigate this we also used MLE to try and estimate both α and ρ at the same time.

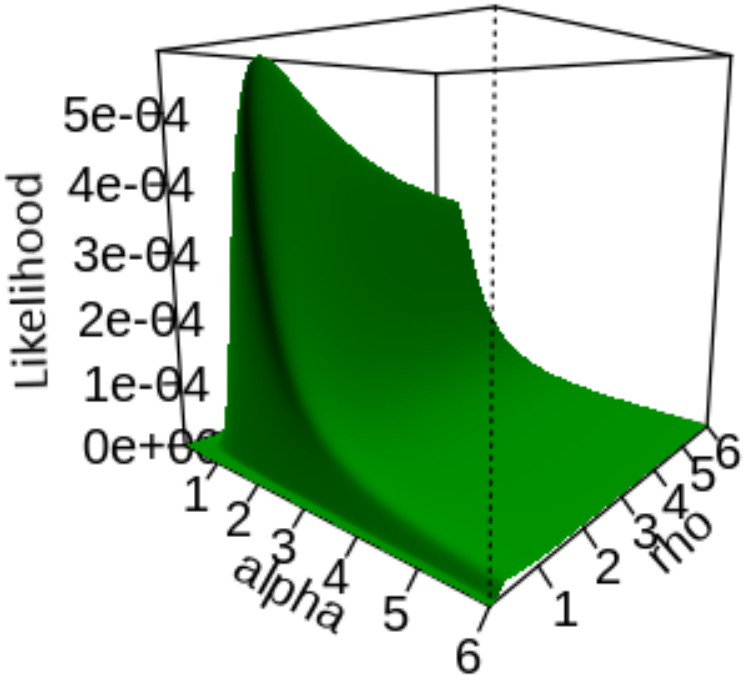


Figure 7: ML estimation of both α and ρ using the Gaussian kernel

This graph estimates α and ρ to be 0.6546 and 1.4216 respectively. However when they are estimated individually they come out to be 0.7372 and 1.1384. So it is reasonable to assume there is a level of co-dependency between α and ρ.

2.5 LOO-CV and MLE distribution

Maximum-likelihood is not the only way of estimating kernel parameters. You could also use LOO-CV which involves removing a sampled point and evaluating how well the prediction created from the remaining points estimates the removed point. Using the data from section 2 we performed LOO-CV to estimate ρ. We found that the optimal $\rho = 1.6458$, which is noticeably worse than the ρ obtained from MLE.

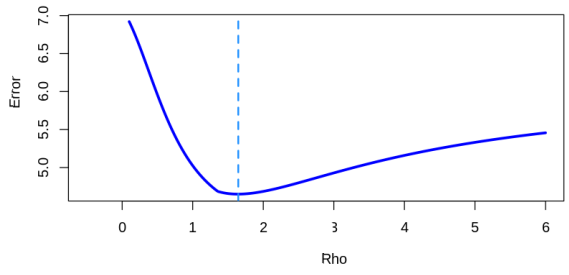
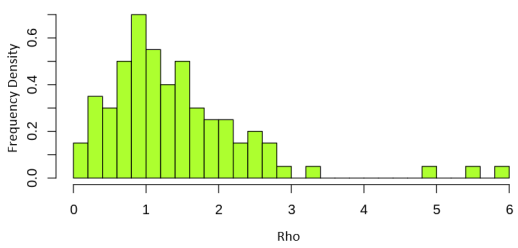
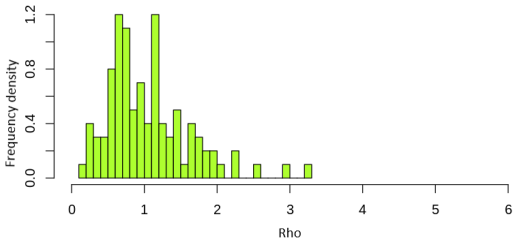


Figure 8: LOO-CV estimation of ρ using a gaussian kernel and 7 points

This large discrepancy between the two methods could be due to the large variance in the ρ obtained from these methods when you only use 7 points. To demonstrate this we have taken random sample points from the initial realisation and we performed MLE on these sampled points. We then repeated this 100 times and plotted the ρs obtained from MLE in the form of a histogram.



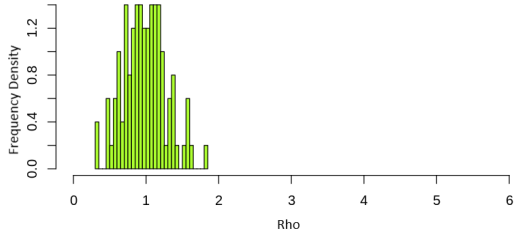
(a) LOO-CV distribution using 7 observations



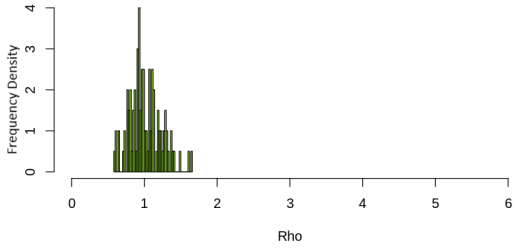
(b) MLE distribution using 7 observations

Figure 9

As you increase the number of points being sampled the variance decreases and the estimators get closer to the true value.



(a) MLE distribution using 20 observations

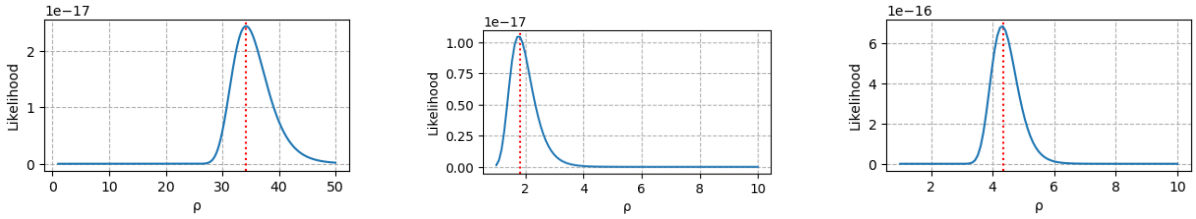


(b) MLE distribution using 40 observations

Figure 10

3 Experiments on the Volcano data

The `DataVolcano50.txt` file contains 50 points $(x_i, y_i, z_i) \in [0,1]^2 \times \mathbb{R}$. First we set the sample vector to be the x and y coordinates of our data and the response vector to be z coordinates. We investigated the use of 3 different kernels on the volcano data set; Gaussian, Laplace, and Matérn-3/2. Our first step was to calculate the most likely value for the kernel parameters ρ for the Kernels using the likelihood function $L(\rho|\mathbf{z} = z) = \dfrac{1}{Det(2\pi K_p)} e^{-\frac{1}{2} z^T K_P^{-1} z}$ where z =our response vector and K_ρ was the kernel matrices made using the kernel functions and the sample locations where K_ρ is dependent on ρ. We plotted the likelihood functions of the ρ parameter for the 3 different kernels.



(a) Likelihood function for the Gaussian kernel

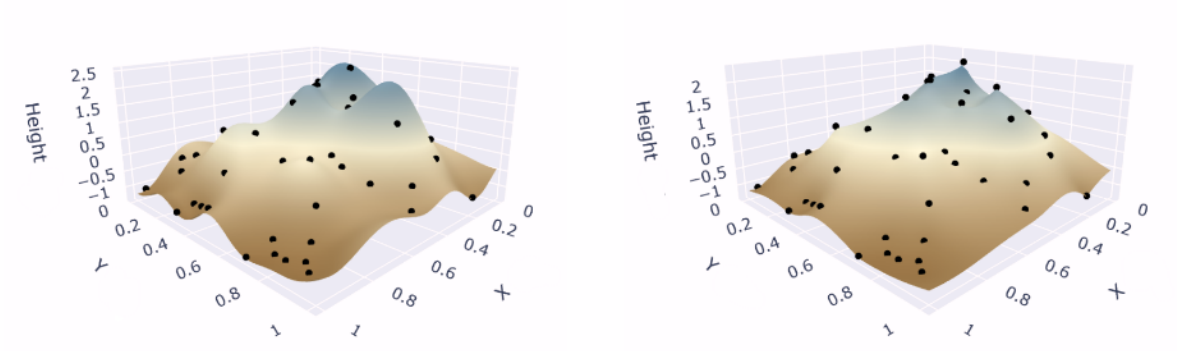
(b) Likelihood function for the Laplace kernel

(c) Likelihood function for the Matérn-3/2 kernel

Figure 11: Likelihood functions

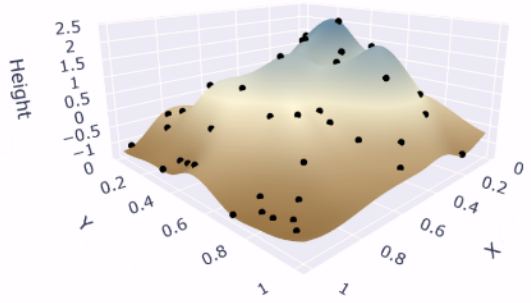
As seen in Figure 11 the approximate values for the rho values were $\rho_{Gaussian} \approx 34.22$, $\rho_{Laplace} \approx 1.81$ and $\rho_{Matérn-3/2} \approx 4.33$.

We then used these parameters in the three kernel functions to obtain the optimum 3D surface plots as seen in Figure 12 by using the formula $\mathbb{E}(Z_x|\mathbf{z}) = \mathbf{k}^T(x)\mathbf{K}^{-1}\mathbf{z}$



(a) Gaussian

(b) Laplace



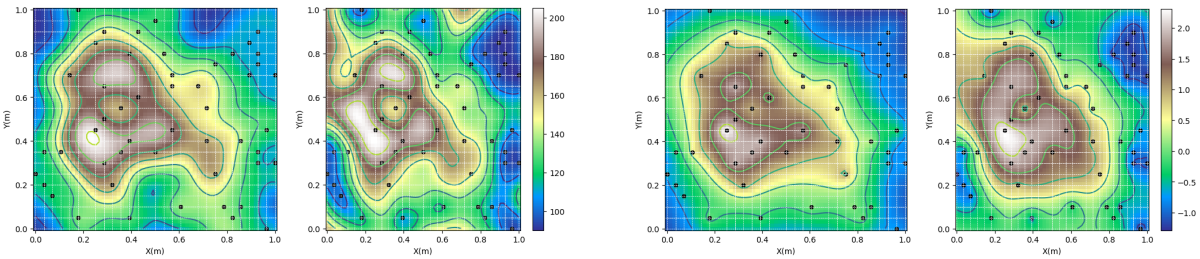
(c) Matérn-3/2

Figure 12: Optimum 3D plots for the surface of Maunga Whau,black dots are the observations.

As seen in figure 12 these plots hold the same properties as those shown in section 2.2 Figure 2 with the Laplace optimum prediction having sharper turns and yielding non-continuously differentiable paths where as the Matérn-3/2 and gaussian optimum predictions have smoother curves and the gaussian curves being infinitely differentiable.

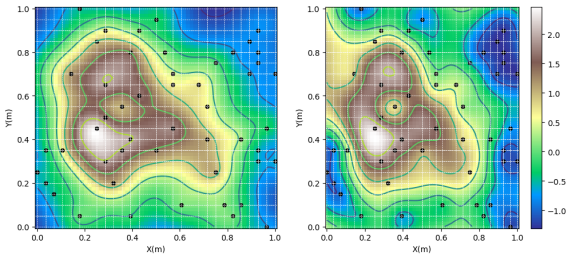
We also used the sample data and kernel functions to produce the upper and lower 95% confidence intervals and plotted them on 2D contour maps as seen in Figure 13 by using the

conditional covariance formula $c - cov(Z_x, Z_{x'}|z) = K(x, x') - \mathbf{k}(x)^\top \mathbf{K}^{-1}\mathbf{k}(x')$.



(a) Gaussian

(b) Laplace



(c) Matérn-3/2

Figure 13: Contour graphs of the 3 plots lower 95% confidence intervals on the left and upper 95% confidence interval on the right, black dots are the observations.

As seen on figure 13 there are more noticeable differences seen by using different kernels for the data analysis of the same data set. On the Laplace figure again we see more sharper curves around the observation points and we can also see that the Gaussian and Matérn-3/2 have quite similar plots though there are some noticeable differences, like how the confidence intervals of the Gaussian figures have more range than that of Matérn-3/2.

4 Observation Noise and Finite-Dimensional Kernels

In addition to the core experiments, we explored two extensions that allow us to visualise kernel regression on a deeper level.

4.1 Models with observation noise

The standard kernel regression model assumes perfect noise-free observations. However, in the real-world, measurements are rarely noise free and often carry independent Gaussian noise. To help simulate a real-world scenario, replaced our kernel matrix $\mathbf{K}$ with $\mathbf{K} + \sigma^2\mathbf{I}$, where $\sigma^2 > 0$ is the noise variance. This modification appears in the prediction formula as

$$\hat{z}(x) = \mathbf{k}(x)^\top (\mathbf{K} + \sigma^2\mathbf{I})^{-1}\mathbf{z}$$

and in the conditional-covariance formula accordingly.

We found as σ^2 increases, the predictor becomes visibly smoother and the confidence bands widen. This behaviour relates directly to the kernel-choice experiments in task (b): a large noise variance has a similar regularising effect to choosing a very smooth kernel or a small ρ.

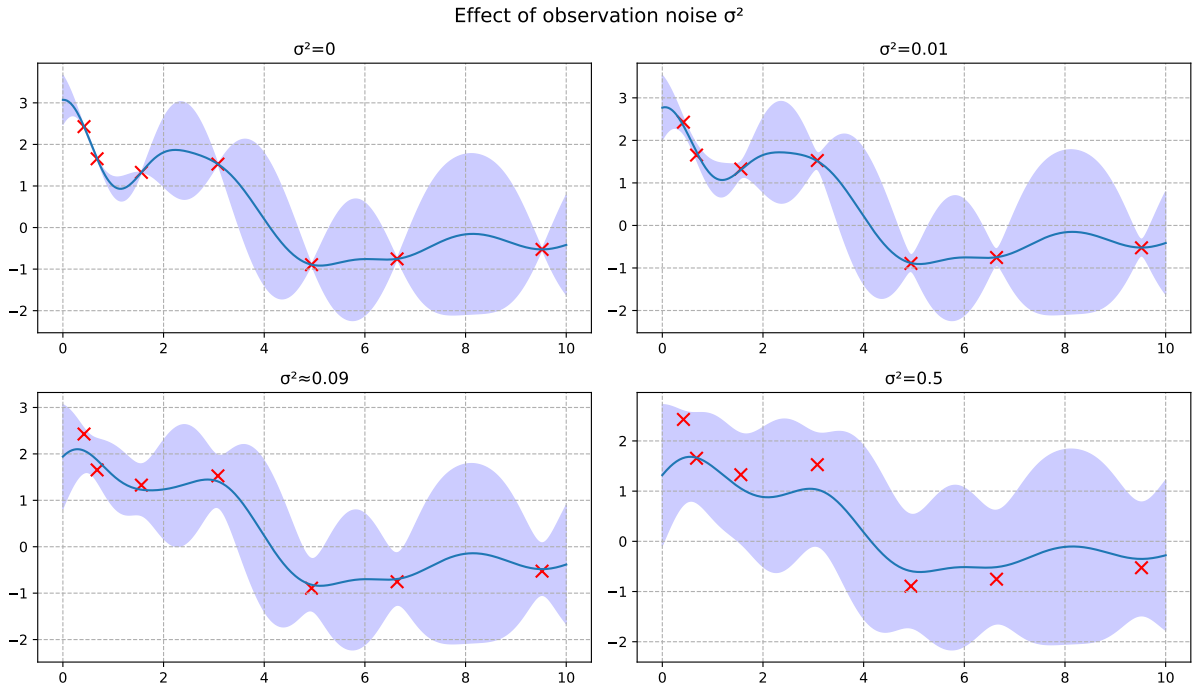


Figure 14: Effect of observation noise σ^2

As you can see when you have a larger noise coefficient the model does not strictly follow the points and as such the confidence intervals around the points widen considerably.

We also applied the noisy model to the Volcano dataset provided. The ML noise estimate was small ($\hat{\sigma}^2 \approx 0.014$), indicating that the original 50 observations are relatively clean. Nevertheless, the noisy predictor is slightly smoother near isolated points, demonstrating that even modest observation noise assists on preventing overfitting in our plot.

4.2 Kernel regression with finite-dimensional kernels

We also implemented finite-dimensional kernels, which are constructed from an explicit basis of functions $g_1(\mathbf{x}), \ldots, g_p(\mathbf{x})$ via

$$K(\mathbf{x}, \mathbf{x}') = \sum_{k=1}^{p} g_k(\mathbf{x}) g_k(\mathbf{x}') = \mathbf{g}(\mathbf{x})^\top \mathbf{g}(\mathbf{x}').$$

The resulting kernel matrix is simply $\mathbf{K} = \mathbf{G}\mathbf{G}^\top$, where $\mathbf{G}$ is the design matrix. This formulation is equivalent to ordinary linear regression in the feature space defined by the basis functions.

For the one-dimensional case we used polynomial bases of increasing degree $(1, x, x^2, \ldots, x^d)$. Higher-degree polynomials produce more flexible (and eventually wavy) predictions, but the uncertainty grows rapidly away from the data. This behavior contrasts with the stationary kernels studied in task (b), which provide local smoothness controlled by ρ.

On the Volcano dataset we used the six basis functions

$$g_1 = 1,\ g_2 = x,\ g_3 = y,\ g_4 = x^2,\ g_5 = y^2,\ g_6 = xy$$

(degree-2 polynomial) and compared degrees 1, 2 and 3. The finite-dimensional models yield globally smooth surfaces that capture the overall shape of the volcano, but however lack the

local detail provided by the Gaussian kernel. Computationally they are cheaper since the kernel matrix is low-rank, but they cannot adapt locally like the nonparametric kernels used in section 2.

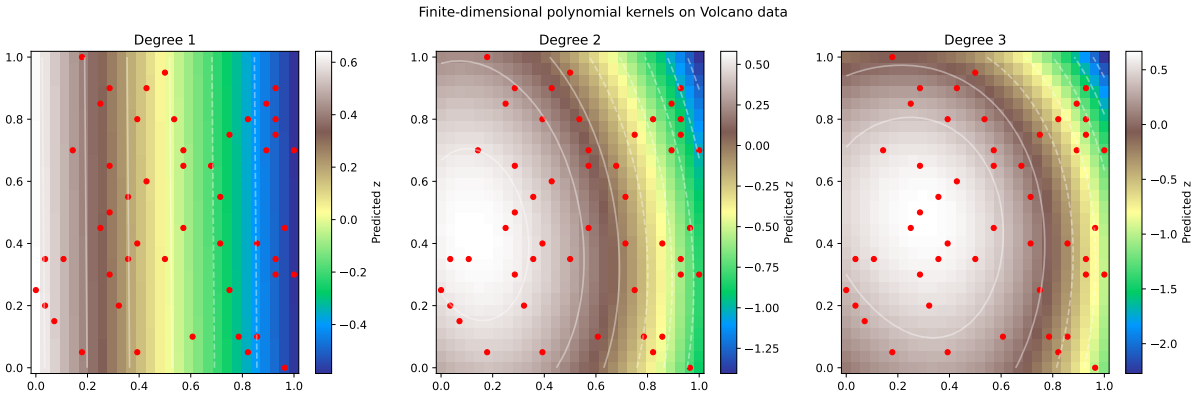


Figure 15: Finite-dimensional polynomial kernels on Volcano data(red dots are the observations).

Finite-dimensional kernels provide a bridge between classical linear regression and the fully nonparametric approach of the methods used in section 2, making the connection to the lecture notes on linear models explicit.

5 Conclusion

We successfully implemented kernel regression from scratch and demonstrated its behavior on both synthetic one-dimensional data and the real volcano dataset. The choice of kernel and the value of the parameter ρ affects the smoothness of the prediction. Maximum-likelihood estimation provides us with a practical way to select ρ, with LOO-CV being an alternative methods that can also estimate which kernel is best for making a prediction. While the addition of an observation-noise term lets us simulate how data would behave in real situations. Finite-dimensional kernels offer a simple, computationally cheap alternative but lose the local flexibility of stationary covariance kernels.